

The Energy Balance Project

Mathematical Foundation Note — Version 2.7

Variational derivation of the local actor law, and a continuous-time energy–dissipation theorem

Status: mathematical specification only. No change is made to the physical engine or the EBU accounting implementation. Claims are labelled by epistemic status (Theorem / Proof sketch / Numerical observation / Regression check / Conjecture). The companion `test_math.py` (8 groups, 34 numerical regression checks) *validates* the derivations at tested points; a passing run is not a proof, and the check count is not a count of theorems.

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1. Two clearly separated models

1.1 Model D — the discrete map the engine runs

State $x = (x_i)$, $x_i \in [0, K_i]$, on a lattice with von Neumann neighbourhood. One tick is the composition $T = A(N(x))$ (apply N then A) of an on-site natural map and an actor transport map. The homeostatic penalty and its marginal:

$$\ell_i(x) = \alpha_i [L_i - x]_+^2 + \beta_i [x - U_i]_+^2, \quad \mu_i = \ell'_i(x) = -2\alpha_i(L_i - x)_+ + 2\beta_i(x - U_i)_+$$

Natural map (order: inflow, regeneration, demand, leak, clip to $[0, K_i]$):

$$(Nx)_i = \Pi(x_i + s_i + g_i(x_i) - d_i - \lambda_i - \kappa_i x_i)$$

Actor map on $y = Nx$: each actor picks the steepest feasible out-edge by the engine force, then sizes the transfer by the raw flux (*gradient* mode) or the line search (*safe* mode); accepted transfers are applied **sequentially** against the live state (load-bearing for §6):

$$F_{ij} = \mu_i - \mu_j - \theta \quad q_{\text{safe}} = \arg \min_{0 \leq q \leq q_{hi}} [\ell_i(y_i - q) + \ell_j(y_j + \eta q)]$$

1.2 Model C — continuous-time approximation (labelled as such)

Reading Model D as forward Euler with $\Delta t = 1$, the $\Delta t \rightarrow 0$ limit is

$$\dot{x}_i = s_i + g_i(x_i) - d_i - \lambda_i - \kappa_i x_i + \sum_{j \in N(i)} (\eta_{ji} J_{ji} - J_{ij})$$

Model C is an approximation, not the engine. The flip bifurcation (§3) and transport overshoot (§5) are $\Delta t = 1$ artifacts absent from Model C; engine-behaviour results use Model D.

2. State functional, dissipation potential, derived law

Keep the **state** functional separate from transport **dissipation**; only the state functional takes cell-state arguments. Transport is not placed inside a state functional (no edge-state variable exists).

$$V_{\text{state}}(x) = B_{\text{hom}} + B_{\text{reg}} = \sum_i \ell_i(x_i) + \sum_{i: \rho_i > 0, A_i > 0} \chi_i [R_i - x_i]_+^2, \quad R_i = A_i + \delta_i$$

$$\Psi(J) = \sum_{e \in E} \left[\frac{J_e^2}{2M_e} + \theta_e J_e \right], \quad J_e \geq 0$$

Local chemical potential (state marginal), on-site by construction:

$$\mu_i^{\text{tot}} = \frac{\partial V_{\text{state}}}{\partial x_i} = \ell'_i(x_i) + r'_i(x_i), \quad r'_i(x) = -2\chi_i(R_i - x) \quad (x < R_i)$$

For a directed edge $e=(i \rightarrow j)$ a flux removes J_e from i and deposits $\eta_e J_e$ at j (lossy continuity). Hence the η -weighted thermodynamic force and the Onsager/gradient-flow flux:

$$\frac{dV_{\text{state}}}{dt} = - \sum_e J_e f_e, \quad f_e := \mu_i^{\text{tot}} - \eta_e \mu_j^{\text{tot}}; \quad J_e = M_e [f_e - \theta_e]_+$$

Derived local law. Edge force $F_e = \mu_i^{\text{tot}} - \eta_e \mu_j^{\text{tot}} - \theta_e$. It reads only x_i , adjacent x_j , and cell/edge-local constants. The engine force $\mu_i - \mu_j - \theta$ matches it **only when** $\eta_e = 1$ and $\chi = 0$; for $\eta_e < 1$ the two differ by $(1 - \eta_e)\mu_j$.

2.1 Three distinct laws (do not conflate)

(1) Continuous Onsager law:

$$J_e = M_e [f_e - \theta_e]_+$$

(2) Raw discrete explicit-Euler rule:

$$q_e = \Delta t M_e [f_e - \theta_e]_+$$

(3) Current safe rule (gated coordinate descent):

$$q_e^* = \arg \min_{0 \leq q \leq q_{hi}} V_{\text{state}}(x + S_e q)$$

Law (3) minimises the *state functional* along the edge; it **excludes** the linear cost $\theta_e q$ and the quadratic dissipation $q^2 / (2M_e)$ of the Onsager objective (θ enters (3) only via edge eligibility; M not at all). So (3) is **gated coordinate descent, not the Onsager flux** — different from (1)/(2) even at $\eta = 1$.

3. Logistic harvest — existence is not sustainability

Isolated logistic source with constant net harvest h ; interior of Model D:

$$h^* = \frac{\rho K}{4}, \quad D = 1 - \frac{4h}{\rho K}, \quad x_{\pm}(h) = \frac{K}{2}(1 \pm \sqrt{D}), \quad \phi'(x_{\pm}) = 1 \mp \rho\sqrt{D}$$

Theorem 3 (corrected). $h \leq \rho K/4$ gives *existence* of equilibria, not sustainability. Persistence needs all three: (i) $h \leq \rho K/4$; (ii) basin $x(0) > x_{-}(h)$ (the always-unstable lower root is the collapse threshold); (iii) discrete stability $\rho\sqrt{D} < 2$ (else a flip/period-doubling bifurcation, equilibrium not attracting).

Numerical observation ($\rho=0.4$, $K=20$): equilibrium matches $x_{+}(h)$ to 4 dp; start just below x_{-} collapses; $\rho=2.6$ ($\rho\sqrt{D}=2.18$) gives a sustained 2-cycle.

4. Allee reserve under driving — the reserve is not $x = A$

For the isolated *undriven* Allee source the basin boundary is $x = A$. Under drive it becomes the unstable middle root x_r of

$$G(x) = \rho x(1 - \frac{x}{K})(\frac{x}{A} - 1) + s - d - \lambda - \kappa x - h = 0$$

$$\frac{\partial x_r}{\partial h} = \frac{\partial x_r}{\partial d} = \frac{\partial x_r}{\partial \lambda} = \frac{1}{G'(x_r)} > 0, \quad \frac{\partial x_r}{\partial K} = \frac{x_r}{G'(x_r)} > 0, \quad \frac{\partial x_r}{\partial s} = -\frac{1}{G'(x_r)} < 0$$

Harvest, demand and leakage **raise** the reserve; supply **lowers** it; transport out acts as harvest, transport in as supply. *Verified* ($\rho=0.6$, $K=20$, $A=5$): $h=0.5 \rightarrow 5.99$, $\kappa=0.05 \rightarrow 5.58$ (both $>A$); $s=0.5 \rightarrow 3.58$ ($<A$).

5. Descent condition — corrected for loss and curvature

Along the lossy transfer, the pair burden $\psi(q) = \alpha_i(y_i - q) + \alpha_j(y_j + \eta q)$ is convex. In the operating branch (source in excess, destination in deficit):

$$\psi'' = 2(\beta_{src} + \eta^2 \alpha_{dst}) \implies M \leq \frac{1}{\max(\alpha_i, \beta_i) + \eta^2 \max(\alpha_j, \beta_j)}$$

The loss enters as η^2 on the destination (earlier $2(\alpha+\beta)$ was the $\eta=1$ case). For symmetric weights the bound is tight: $M \leq 1/(w(1+\eta^2))$. This governs the **raw** rule (2) only; the safe line search (3) has no

mobility bound (size set by q_{hi} and the acceptance gate).

Validation: symmetric weights hit the bound exactly (e.g. $w=1, \eta=0.9 \rightarrow 0.5525$); asymmetric weights make it conservative (safe).

6. Finite causal speed — corrected for sequential execution

Theorem 6.1. If all edge fluxes are computed from the frozen state $y = Nx$ and applied **simultaneously**, then $x_i(t+n)$ depends only on cells within graph distance n : causal speed ≤ 1 cell/tick. *Validated:* frozen-state leak = 0.

Observation 6.2: the current engine freezes proposal *directions* but applies accepted transfers **sequentially** against live state, so a chain $0 \rightarrow 1 \rightarrow 2$ lets a perturbation at cell 0 reach cell 2 in one tick (measured +0.124). We adopt the honest weakened statement for the current engine (≤ 1 edge per micro-step; per tick bounded by the longest accepted-transfer chain) and record strict one-hop (frozen-state simultaneous update) as the target discipline. No engine change is made.

7. Verdict: A, B_raw, B_safe, and discretisation (C)

A — derived Onsager flux $J_e = M_e[\mu_i^{\text{tot}} - \eta_e \mu_j^{\text{tot}} - \theta_e]_+$. **B_raw** — engine raw rule (loss-blind direction, size $M[F]_+$, $\chi=0$ default, sequential). **B_safe** — engine safe rule (same direction, size $\argmin V_{\text{state}}$, sequential).

C — when B_raw is the forward-Euler discretisation of A. This is a statement about the *scheme*, not the trajectories: even under all conditions the finite- Δt discrete trajectory is **not** identical to the continuous flow of A — forward Euler has local truncation error $O(\Delta t^2)$ (global $O(\Delta t)$), and the engine runs at $\Delta t = 1$. Conditions: (1) $\eta_e = 1$; (2) $\chi = 0$ on active edges; (3) $c_0 = 0$ (a fixed activation cost is a discrete jump $\rightarrow$ hybrid); (4) no overlapping sequential transfers; (5) mobility bound (§5).

A $\neq$ B_safe in general, even at $\eta=1, \chi=0, c_0=0$, single transfer: B_safe minimises the state functional (exact minimiser), not the linear flux $M_e F_e$. It is a different (coordinate-descent) law.

7.1 C-1 promoted: continuous-time energy–dissipation theorem

Theorem 7.1. Assume continuous time (Model C); $c_0 = 0$; fixed graph; local lossy continuity with $\eta_e \in (0, 1]$; Onsager flux. Each penalty is C^1 , so $V_{\text{state}} \in C^1$ and the RHS is Lipschitz (unique C^1 solutions). With drive $u_i = s_i + g_i(x_i) - d_i - \lambda_i - \kappa_i x_i$:

$$\frac{dV_{\text{state}}}{dt} = \sum_i \mu_i u_i - \sum_e \left[\frac{J_e^2}{M_e} + \theta_e J_e \right] \leq \sum_i \mu_i u_i$$

Proof. Continuity gives $dx_i/dt = u_i + \sum \text{incoming } \eta J - \sum \text{outgoing } J$. By the chain rule $dV/dt = \sum_i \mu_i (dx_i/dt) = \sum_i \mu_i u_i + T$. Reorganising T by edge, edge $(i \rightarrow j)$ contributes $-J_e(\mu_i - \eta_e \mu_j) = -J_e f_e$, so $T = -\sum_e f_e J_e$. On an active edge $J_e = M_e(f_e - \theta_e)$, hence $f_e J_e = J_e^2/M_e + \theta_e J_e$; on an inactive edge both sides vanish. Summing gives the identity; dropping the non-negative dissipation gives the inequality. ■

Excluded engine mechanisms (scope of Theorem 7.1). The theorem covers the smooth, unconstrained, continuous-time flow only. It does **not** cover, and each needs its own nonsmooth / projected / hybrid / discrete analysis: clipping/projection at 0 and K ; spill at K ; unmet-demand saturation; hard-reserve constraints; the fixed activation cost c_0 ; sequential live-state transfers (§6); and finite-step ($\Delta t = 1$) updates (an $O(\Delta t^2)$ remainder).

Corollary 7.2 (undriven, LaSalle, stated precisely). If $u_i \equiv 0$ and trajectories are bounded (they are, $x \in [0, K]$), then V_{state} is non-increasing and by LaSalle the flow converges to the **largest invariant subset** of the zero-dissipation set $Z = \{ J_e = 0 \ \forall e \} = \{ f_e \leq \theta_e \ \forall e \}$. This

claims **only** convergence to Z . It does **not** assert $V_{\text{state}} = 0$, complete homeostasis, a unique equilibrium, or which point of Z is reached — Z is generally a continuum, and any stronger claim needs a separate argument.

Numerical validation of Theorem 7.1: fine- Δt integration matched the identity to $O(\Delta t)$ (max residual 6.2×10^{-5} at $\Delta t = 10^{-4}$); dV/dt turned positive once the drive exceeded the dissipation, consistent with the inequality biting only when the drive is non-positive.

8. Open conjectures and regression validation

Conjecture C-1' (discrete driven bound). A discrete counterpart of the identity holds for B_{raw} under the mobility bound: $V(x_{\{t+1\}}) - V(x_t) \leq \sum_i \mu_i u_i - D_t$ with $D_t \geq 0$ (the continuous case is now Theorem 7.1; the discrete case remains open pending the step-size remainder and the §6 sequential-execution caveat).

Conjecture C-2 (reserve surrogate). V2.4's constant $R_i = A_i + \delta_i$ dominates the driven reserve x_r (hence is safe) iff $\delta_i \geq x_r - A_i$ over the operating envelope.

Regression validation. `test_math.py` contains 8 groups holding 34 numerical regression checks. The check count is a harness detail, not a count of theorems; a passing run validates the tested points only.

Group	Guards
1. fold & basin	equilibrium = $x_+(h)$; basin straddle of $x_-(h)$ (Thm 3)
2. flip	2-cycle appears once $\rho\sqrt{D} > 2$ (Thm 3)
3. driven reserve	middle root moves in the signed directions (Thm 4)
4. descent bound	$M \leq 1/[\max(\alpha, \beta)_i + \eta^2 \max(\alpha, \beta)_j]$ safe; tight if symmetric
5. force gap	engine vs derived force coincide iff $\eta = 1$
6. causality	sequential leak > 0 ; frozen-state simultaneous leak $= 0$
7. energy identity	dV/dt identity residual is $O(\Delta t)$ (Thm 7.1)
8. three-law split	$q_{\text{safe}} \neq M \cdot F$ even at $\eta=1, \theta=0$

Corrections applied in this note (relative to the first informal draft): logistic existence $\neq$ sustainability; the Allee reserve shifts off A under drive; the descent bound carries an η^2 loss factor and uses the max-over-branch curvature; the derived force is loss-weighted ($\mu_i - \eta\mu_j - \theta$); transport is a dissipation potential, not a state term; one-hop causality holds only under frozen-state simultaneous updates; the engine splits into B_{raw} and B_{safe} ; and C-1 is now a continuous-time theorem.